MACHINE LEARNING FOR BREAST CANCER OUTCOME PREDICTION

Predicting pathological complete response (PCR) and relapse-free survival (RFS) using multimodal clinical and MRI-derived data

AUTHOR	PROJECT DATE	DATA SOURCE
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PROJECT REPORT

A comparative machine learning study covering data preparation, feature selection, regression, classification and model evaluation.

This report presents the methodology and results documented in the original project paper, with clearer structure, expanded technical explanation and explicit notes where the source does not provide enough detail to draw further conclusions.

Executive summary

This project investigated whether machine learning models could use multimodal information from the I-SPY 2 TRIAL to predict two breast cancer treatment outcomes: pathological complete response (PCR) and relapse-free survival (RFS). The dataset was described as containing 10 clinical features and 107 MRI-based features. Two modelling tasks were developed: a classification task for PCR and a regression task for RFS.

The workflow combined missing-value imputation, feature selection, scaling and model comparison. For RFS, three non-linear regression approaches were compared alongside a linear regression baseline. Support Vector Regression (SVR) produced the lowest reported cross-validation MAE among the non-baseline models (0.145), narrowly ahead of Random Forest (0.148) and Decision Tree (0.154). For PCR, Random Forest achieved the highest reported validation balanced accuracy (0.90), closely followed by SVM (0.89).

Task	Target	Best reported model	Reported result
Regression	Relapse-free survival (RFS)	Support Vector Regressor	MAE 0.145
Classification	Pathological complete response (PCR)	Random Forest Classifier	Balanced accuracy 0.90

Key interpretation

The reported results suggest that different algorithms performed best for the two prediction tasks: SVR for RFS regression and Random Forest for PCR classification. The classification models produced relatively strong validation balanced-accuracy scores in the reported experiments. The regression results require more cautious interpretation because the linear-regression baseline MAE (59.5) is on a markedly different scale from the other reported MAE values; the original report does not explain this discrepancy.

Scope and intended use

This is an academic machine learning project, not a clinically validated decision-support system. The results should be interpreted as model-comparison findings on the project dataset. They do not establish clinical effectiveness, patient benefit, or readiness for deployment in healthcare.

1. Project context and objectives

Breast cancer treatment response varies between patients, creating an opportunity for predictive modelling to support research into treatment outcomes. This project examined two distinct outcome measures. PCR is treated as a binary classification target, while RFS is treated as a continuous regression target. Using both clinical and MRI-derived variables enabled the study to explore whether complementary data modalities could contribute to outcome prediction.

1.1 Research objectives

The project had four practical objectives:

1. Prepare the supplied dataset for supervised learning by addressing missing values and scaling numeric features.
2. Reduce the feature space to limit redundant or less informative inputs.
3. Train and tune multiple regression and classification algorithms using cross-validation and held-out data splits.
4. Compare model performance using metrics suited to each task: MAE for RFS and balanced accuracy for PCR.

1.2 Dataset overview

Dataset characteristic	Description recorded in the project
Source	I-SPY 2 TRIAL
Input modalities	Clinical variables and MRI-derived variables
Clinical features	10
MRI-based features	107
Prediction targets	PCR (classification) and RFS (regression)

The source report does not state the total number of patients or observations, the precise definitions and units of every input feature, the prevalence of each outcome class, or the exact target encoding. Those details are therefore not inferred in this revised report.

1.3 Overall workflow

The project followed a conventional supervised-learning pipeline: dataset preparation → missing-value treatment → feature selection → scaling and data partitioning → model training and hyperparameter search → validation and cross-validation → comparison of reported metrics. Separate preprocessing and modelling workflows were used for the regression and classification tasks because their targets and evaluation requirements differ.

2. Regression: predicting relapse-free survival

The regression workflow aimed to estimate RFS from the available input features. The target was separated from the predictors, and the data was prepared for training and evaluation. The project compared a simple linear baseline with Decision Tree, Support Vector Machine and Random Forest regressors.

2.1 Data preparation

Missing-value imputation. Missing entries were first represented as NaN values and then imputed using a K-Nearest Neighbours (KNN) approach. A subsequent check was performed to confirm that no missing values remained. KNN imputation estimates missing values using observations that are similar across the available feature space.

Correlation-based feature reduction. A correlation matrix was calculated for numeric columns, excluding PCR and RFS. Feature pairs with an absolute correlation threshold of 0.8 were identified. Where two features were highly correlated, the feature with the weaker relationship to the target was removed. This was intended to reduce redundancy and limit multicollinearity while retaining target-relevant information.

Train/test split and scaling. The RFS dataset was divided into 80% training data and 20% test data. Min-Max scaling was applied to numeric features to place them on a common scale. This is particularly relevant for distance- and margin-based methods, where features with larger numeric ranges can otherwise dominate model fitting.

2.2 Models and reported configuration

Model	Reported configuration / evaluation	Reported result
Linear Regression	Baseline model; evaluated using MSE, R^2 and MAE	MAE 59.5
Decision Tree Regressor	Max depth = 1; min samples split = 3; min samples leaf = 5; 5-fold CV	MAE 0.154
Support Vector Regressor	C = 0.1; gamma = 'auto'; epsilon = 0.1; grid search and k-fold CV	MAE 0.145
Random Forest Regressor	Max depth = 5; min samples split = 2; 50 estimators; 5-fold CV	MAE 0.148

Linear Regression baseline. Linear Regression was trained on the training partition and evaluated on the test set. The original report records MSE, R-squared and MAE as evaluation measures, and gives a baseline MAE of 59.5. It does not provide the corresponding MSE or R-squared values.

Decision Tree Regressor. Grid search was used to select the reported tree configuration. Five-fold cross-validation was then used to estimate performance across different data partitions. The reported average MAE was 0.154.

Support Vector Regressor. The SVR model used the reported hyperparameters C = 0.1, gamma = 'auto' and epsilon = 0.1. Following grid search, cross-validation produced a mean MAE of 0.145, the lowest of the three non-baseline regression models.

Random Forest Regressor. The selected configuration used 50 estimators, maximum depth 5 and minimum split size 2. Five-fold cross-validation produced a mean MAE of 0.148.

3. Regression results and interpretation

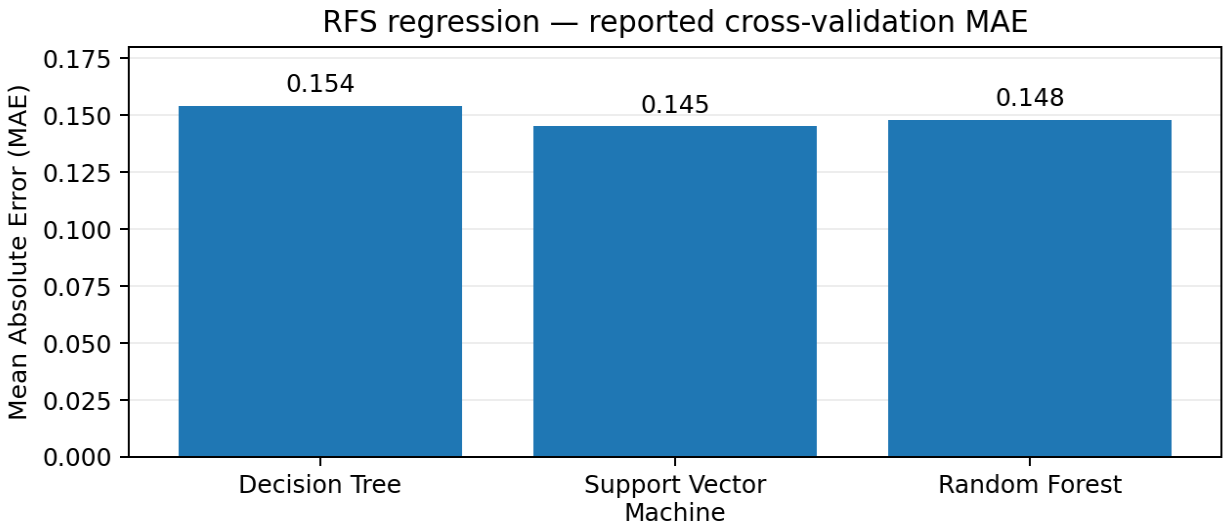


Figure 1. Comparison of the reported cross-validation MAE values for the three non-baseline regression models. Lower MAE indicates a smaller average absolute prediction error, provided the scores are measured on the same target scale and evaluation basis.

3.1 Comparative findings

Model	Reported MAE	Relative position
Support Vector Regressor	0.145	Lowest reported MAE
Random Forest Regressor	0.148	0.003 higher than SVR
Decision Tree Regressor	0.154	0.009 higher than SVR

The SVR model marginally outperformed the other two non-baseline models. The difference between SVR and Random Forest was 0.003 MAE, while the difference between SVR and Decision Tree was 0.009. These are small numerical gaps, so the results support a cautious conclusion that SVR was the strongest of the reported regression models rather than demonstrating a substantial performance advantage.

3.2 Metric selection

Mean Absolute Error measures the average absolute difference between predicted and observed values. It is expressed in the same units as the target variable and is relatively straightforward to interpret. Unlike squared-error metrics, MAE does not disproportionately increase the contribution of larger individual errors. The original report therefore selected MAE as the primary measure for comparing RFS regression models.

3.3 Important reporting caveat

The reported Linear Regression baseline MAE of 59.5 is not directly comparable with the cross-validation MAE values of 0.145–0.154 without further information. The original paper does not explain whether the baseline was evaluated on a different target scale, used a different split, or was calculated under a different preprocessing procedure. This revised report preserves the value as originally stated and does not attempt to correct it. A portfolio reviewer should regard the baseline comparison as unresolved until the original implementation or evaluation output can be checked.

4. Classification: predicting pathological complete response

The classification workflow aimed to predict PCR as a binary outcome. Because class imbalance can cause a model to favour the more frequent class, the project used feature selection, outlier treatment and minority-class upsampling before training and evaluation. Balanced accuracy was selected as the principal comparison metric.

4.1 Data preparation

Missing-value imputation. The classification workflow followed the same KNN-based imputation process described for regression: missing values were represented as NaN, imputed and checked.

ANOVA F-test feature selection. The project used an ANOVA F-test scoring function to select the 25 features considered most relevant to the binary PCR target. Reducing the feature set in this way was intended to focus model training on a smaller group of potentially informative predictors.

Winsorisation. Each of the selected 25 features was Winsorised using a 5% limit at both the lower and upper ends. Winsorisation limits the influence of extreme values by capping observations at selected percentile boundaries rather than deleting them.

Class balancing. The majority and minority classes were separated, and the minority class was upsampled until its sample count matched the majority class. The two groups were then recombined to create a balanced dataset for the subsequent workflow.

Data split and normalization. The data was divided into training (70%), validation (15%) and test (15%) partitions. Stratification was used to preserve class proportions across the splits. Min-Max scaling was applied to the feature sets in preparation for model training.

4.2 Models and reported results

Model	Reported configuration	Validation balanced accuracy
Logistic Regression	Baseline classifier	0.70
Decision Tree	Max depth = None; min samples leaf = 1; min samples split = 2; 5-fold CV	0.77
Support Vector Machine	C = 10; gamma = 1; RBF kernel; stratified k-fold grid search	0.89
Random Forest	200 estimators; max depth = None; min samples split = 2; min samples leaf = 1	0.90

Logistic Regression. The baseline classifier was trained on the training set and used to generate predictions for the training, validation and test sets. The reported validation balanced accuracy was 0.70.

Decision Tree. Grid search with five-fold cross-validation was used to tune the classifier. The reported configuration had no maximum depth restriction, a minimum leaf size of 1 and a minimum split size of 2. Validation balanced accuracy was 0.77.

5. Classification results and interpretation

Support Vector Machine. A parameter grid was evaluated using stratified k-fold cross-validation. The selected configuration used C = 10, gamma = 1 and an RBF kernel. The validation balanced accuracy was 0.89.

Random Forest. Grid search with cross-validation selected 200 estimators, no maximum depth restriction, minimum split size 2 and minimum leaf size 1. The validation balanced accuracy was 0.90, the highest reported score in the classification comparison.

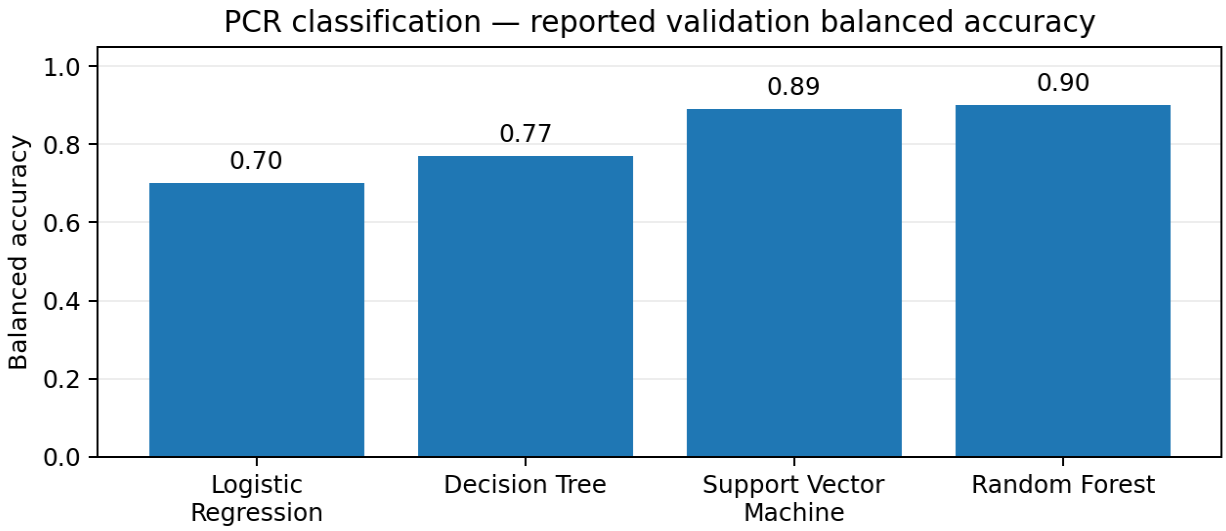


Figure 2. Reported validation balanced accuracy for the four PCR classification models. Higher balanced accuracy indicates better average recall across the classes.

5.1 Comparative findings

Random Forest achieved the highest reported validation balanced accuracy at 0.90, with SVM close behind at 0.89. The Decision Tree reached 0.77, while Logistic Regression recorded 0.70. The results indicate that the two ensemble/kernel-based approaches performed better than the simpler baseline and single-tree classifier in this experiment.

Balanced accuracy is calculated as the average recall across the classes. It is useful when class frequencies differ because it gives each class equal weight in the final score. This makes it more informative than ordinary accuracy when a classifier could otherwise achieve a seemingly high score by predicting the majority class frequently.

5.2 Overall conclusion

Across the two prediction tasks, the strongest reported model differed by target. Support Vector Regression achieved the lowest reported cross-validation MAE for RFS (0.145), while Random Forest achieved the highest reported validation balanced accuracy for PCR (0.90). The classification results show a narrow lead for Random Forest over SVM, suggesting that both models were competitive under the reported evaluation procedure.

The project demonstrates an end-to-end application of supervised machine learning to a multimodal biomedical dataset. It combines data cleaning, feature reduction, outlier handling, class balancing, scaling, hyperparameter tuning and task-specific evaluation. The comparison also illustrates why model selection should be based on the objective and metric of the prediction task rather than assuming that one algorithm will perform best in every setting.

6. Methodological considerations and limitations

A professional interpretation of machine learning results requires attention not only to the highest score, but also to how the evaluation was designed and what information is available to reproduce it. The original report provides a useful outline of the modelling pipeline, but several implementation details are not documented in sufficient depth to independently verify the results.

6.1 Information not specified in the source report

The following details are not provided in the original paper and should be treated as open reporting gaps rather than assumed facts:

- Total dataset size and number of observations in each split.
- Exact feature names, definitions, units and MRI feature extraction procedure.
- PCR and RFS target definitions, encoding and handling of censored survival observations.
- Exact KNN imputation settings, including the number of neighbours and distance metric.
- Whether feature selection, imputation, Winsorisation and scaling were fitted using training data only within each cross-validation fold.
- The complete hyperparameter search spaces and selection criteria.
- Final held-out test-set results for the classification models and the non-baseline regression models.
- Confidence intervals, repeated-run variability, calibration, sensitivity, specificity and confusion matrices.

6.2 Potential data leakage considerations

Preprocessing operations such as imputation, feature selection, Winsorisation, upsampling and scaling should be fitted using training data only, then applied to validation or test data. If these steps are performed before the data is split, or outside the cross-validation folds, information from evaluation data can influence model development and make reported performance appear more favourable than it would be on genuinely unseen observations. The source report does not provide enough procedural detail to confirm exactly how this was handled.

6.3 Evaluation and generalisation

The classification results quoted in the report are validation balanced-accuracy scores. They should not be described as final test-set performance. Similarly, the regression results are described as cross-validation MAE values, while the linear baseline is described as being evaluated on a test set. Comparisons across different evaluation procedures should therefore be made cautiously. An independent external validation dataset would be needed to assess whether the models generalise beyond the data used in this project.

6.4 Clinical interpretation

Predicting PCR or RFS from clinical and imaging variables is a research problem with potentially important implications, but model performance alone is not sufficient for clinical use. A deployable clinical model would require rigorous external validation, assessment of subgroup performance and bias, calibration where appropriate, transparent documentation, clinical oversight and evaluation of whether predictions improve decisions or patient outcomes. No such deployment or clinical validation is claimed here.

7. Skills demonstrated and project summary

This project provides evidence of practical experience across the machine learning lifecycle, from preparing a real-world-style biomedical dataset to comparing models against task-specific performance measures.

Area	Work demonstrated
Data preparation	Missing-value handling using KNN imputation; Min-Max scaling; train/validation/test partitioning.
Feature engineering and selection	Correlation-based feature reduction; ANOVA F-test selection of 25 features; 5% Winsorisation.
Imbalanced learning	Minority-class upsampling to match the majority-class count.
Model development	Linear and Logistic Regression baselines; Decision Tree; Support Vector Machine; Random Forest.
Model optimization	Grid search, hyperparameter selection, k-fold and stratified k-fold cross-validation.
Model evaluation	MAE for RFS regression; balanced accuracy for PCR classification; comparative interpretation.

Portfolio-ready project summary

Designed and evaluated machine learning pipelines to predict breast cancer treatment outcomes using multimodal clinical and MRI-derived data from the I-SPY 2 TRIAL. Implemented preprocessing and feature-selection workflows, addressed class imbalance, and compared regression and classification algorithms using cross-validation and task-appropriate metrics. The strongest reported results were an SVR MAE of 0.145 for RFS and Random Forest balanced accuracy of 0.90 for PCR.

Final takeaway

The main outcome of the project was a comparative analysis showing that model performance depends on the prediction task: Support Vector Regression was strongest for the reported RFS regression experiment, while Random Forest was strongest for PCR classification. The work also highlights the importance of careful preprocessing, suitable evaluation metrics and transparent reporting when applying machine learning to sensitive biomedical problems.

References

Ahmed, F.S. (2023). *Use of machine learning to predict pathological complete response using multi-modal data from I-SPY2 clinical trials.* Journal of Clinical Oncology. The original report does not provide further publication details in the reference entry.

Kornak, J. et al. (2020). *Predicting breast cancer response to neoadjuvant treatment using multi-feature MRI: results from the I-SPY 2 TRIAL.* npj Breast Cancer, 6, 1–6. The original reference entry contains an incomplete author listing in the supplied document; this citation is shortened rather than reconstructed.

Source note: This revised report is based on the supplied five-page project paper. The original contribution-allocation page has been removed, and the author line has been changed to Nderon Beqiri only. Technical values have been retained as reported; explanatory material and limitations have been added for clarity without inventing experimental results.